

Secure Auth Portal (Experiment 7 Version)

This is an advanced Full-Stack mini-project designed for students to understand how modern, secure web applications handle authentication and data protection. It specifically covers the core topics of **Experiment 7: Scalable Read APIs with Caching & Optimization (Now focused on Secure APIs with JWT, RBAC & Encryption)**.

🌟 What is this project about?

This project is a **Secure Dashboard** application that demonstrates enterprise-level backend security patterns.

- **Frontend (FE):** Built with **React** (using Vite). It features a premium, responsive UI with secure routing. It manages Access and Refresh tokens (Token Lifecycle), handles login/logout flows, and dynamically renders UI elements based on the user's Role (RBAC).
- **Backend (BE):** Built with **Spring Boot** (Java) and an **H2 In-Memory Database**. The backend is secured using **Spring Security**. It implements stateless **JWT Authentication**, role-based method security (`@PreAuthorize`), and uses **AES Encryption** to safely store simulated third-party credentials.

How to Run this Project

You will need to run the Backend and the Frontend separately. Open two different terminal windows.

1 Starting the Backend (Spring Boot)

1. Open a terminal.
2. Navigate into the `backend` folder:

```
cd exp_7_code/backend
```

3. Run the Spring Boot application using Maven:

```
mvn spring-boot:run
```

4. The backend will start running on <http://localhost:8080>.
 - Note: You can view the raw database at <http://localhost:8080/h2-console> (JDBC URL: `jdbc:h2:mem:testdb` , Username: `sa` , Password: `<empty>`).
 - Note: The database is automatically seeded with two users for testing:
 - **Admin:** Username: `admin` , Password: `admin123`
 - **User:** Username: `user` , Password: `user123`

2 Starting the Frontend (React)

1. Open a *new* terminal window.
2. Navigate into the `frontend` folder:

```
cd exp_7_code/frontend
```

3. Install the required Node dependencies (you only need to do this once):

```
npm install
```

4. Start the React development server:

```
npm run dev
```

5. Open your browser and go to the URL shown in the terminal (usually <http://localhost:5173>).



Topics Covered in this Code:

- **React (Frontend):**
 - Advanced context/state management for user authentication.
 - Axios interceptors to automatically attach the `Authorization: Bearer <token>` header to outgoing requests.
 - Silent token refresh flows using long-lived refresh tokens.
 - Protected Routes that redirect unauthenticated users to the login page.
- **Spring Boot (Backend) [Experiment 7 Core]:**
 - **Spring Security:** A custom `SecurityFilterChain` to protect all API endpoints except `/api/auth/**`.
 - **JWT Generation & Validation:** Generating signed JWTs on login and parsing them via a custom `OncePerRequestFilter`.
 - **Role-Based Access Control (RBAC):** Using `@PreAuthorize` to restrict access (e.g., distinguishing between `ROLE_USER` and `ROLE_ADMIN`).
 - **AES Encryption:** A utility class that encrypts sensitive simulated data before it is saved to the H2 database, ensuring it is never stored in plaintext.

Enjoy coding securely! 🛡️